Data Visualization Lab — Practical No. 6
Power BI - BLOG

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Practical No. 6

Adventure Works Sales Analysis with DAX Studio and Power BI

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1. Introduction

Numbers sitting in CSV files do not tell a story until they are turned into measures that answer business questions: how much did we sell, how much of it was profit, which products and regions drive the business, and how is it changing over time. In this practical I answer those questions on the Adventure Works dataset by writing custom DAX measures in Power BI Desktop, validating them with DAX Studio, and presenting the results on an interactive dashboard.

This post walks through the whole process in order: the data and model, the KPI measures, the extra DAX calculations, the time-based analysis, the DAX Studio validation, the dashboard, and finally the insights I drew from the results.

Objectives

- Create the core KPI measures: Total Sales, Total Cost, Total Profit, Total Quantity, Total Orders, Total Customers, Average Sales, Average Order Value, Average Product Price and Profit Margin (%).
- Implement at least 20 custom DAX measures using SUM, AVERAGE, COUNT, DISTINCTCOUNT, CALCULATE, DIVIDE, FILTER and RANKX.
- Perform time-based analysis: monthly sales, previous-period sales, sales variance and sales growth percentage.
- Use DAX Studio to run queries, validate the measures and look at query performance.
- Build an interactive dashboard with KPI cards, at least four visuals and slicers for year, product category and region.
- Interpret the results: top products, best categories and regions, sales trend and profitability.

2. Dataset and Data Model

The Adventure Works dataset comes as a set of CSV files: three yearly sales files and several lookup tables. I loaded the files below and left out the unpivot demo file, which is unrelated to this analysis.

Table	Role in the model
Sales (2020, 2021, 2022)	Fact table, 56,046 order lines after appending the three yearly files (OrderDate, OrderNumber, ProductKey, CustomerKey, TerritoryKey, OrderQuantity)
Product	ProductName, ProductCost and ProductPrice
Subcategories and Categories	Product hierarchy used for the category slicer and charts

Table	Role in the model
Customer	Used to count unique customers
Territory	Region and country, used for the region slicer
Calendar	Date table, marked as the date table, with Year, Month Number and Month Name
Returns	Optional table for return quantities

The sales files do not contain a sales amount or a cost column, only the quantity ordered. Sales and cost are therefore calculated row by row: Sales = OrderQuantity × ProductPrice and Cost = OrderQuantity × ProductCost, using RELATED to pull the price and cost from the Product table.

Relationships

All relationships are one-to-many with a single filter direction:

- Calendar[Date] → Sales[OrderDate]
- Product[ProductKey] → Sales[ProductKey]
- Customer[CustomerKey] → Sales[CustomerKey]
- Territory[SalesTerritoryKey] → Sales[TerritoryKey]
- Categories[ProductCategoryKey] → Subcategories[ProductCategoryKey]
- Subcategories[ProductSubcategoryKey] → Product[ProductSubcategoryKey]

On the Calendar table I added three calculated columns and sorted Month Name by Month Number so that months appear in calendar order instead of alphabetically:

```
Year = YEAR(Calendar[Date])
Month Number = MONTH(Calendar[Date])
Month Name = FORMAT(Calendar[Date], "MMM")
```

3. KPI Measures

All measures live in a dedicated table called _Measures so that they are easy to find. The ten measures required by the assignment are listed below.

Sr.	Measure	DAX
1	Total Sales	`SUMX(Sales, Sales[OrderQuantity] * RELATED(Product[ProductPrice]))`
2	Total Cost	`SUMX(Sales, Sales[OrderQuantity] * RELATED(Product[ProductCost]))`
3	Total Profit	`[Total Sales] - [Total Cost]`
4	Total Quantity	`SUM(Sales[OrderQuantity])`
5	Total Orders	`DISTINCTCOUNT(Sales[OrderNumber])`
6	Total Customers	`DISTINCTCOUNT(Sales[CustomerKey])`

Sr.	Measure	DAX
7	Average Sales	`DIVIDE([Total Sales], COUNTROWS(Sales))`
8	Average Order Value	`DIVIDE([Total Sales], [Total Orders])`
9	Average Product Price	`AVERAGE(Product[ProductPrice])`
10	Profit Margin %	`DIVIDE([Total Profit], [Total Sales]) * 100`

Total Orders uses DISTINCTCOUNT on the order number because one order can contain several order lines; counting rows would overstate the number of orders.

4. Additional DAX Calculations

Beyond the ten required measures, I created more calculations that use CALCULATE, FILTER and RANKX, plus the four time-based measures. Together with the KPI measures this gives 20 custom measures.

Sr.	Measure	DAX	Functions
11	Monthly Sales	`CALCULATE([Total Sales], VALUES(Calendar[Month Name]))`	CALCULATE, VALUES
12	Previous Month Sales	`CALCULATE([Total Sales], DATEADD(Calendar[Date], -1, MONTH))`	CALCULATE, DATEADD
13	Sales Variance	`[Total Sales] - [Previous Month Sales]`	Arithmetic
14	Sales Growth %	`DIVIDE([Sales Variance], [Previous Month Sales]) * 100`	DIVIDE
15	Order Line Count	`COUNT(Sales[OrderLineItem])`	COUNT
16	Bikes Sales	`CALCULATE([Total Sales], Categories[CategoryName] = "Bikes")`	CALCULATE
17	High Value Sales	`CALCULATE([Total Sales], FILTER(Product, Product[ProductPrice] > 1000))`	CALCULATE, FILTER
18	Avg Cost per Unit	`DIVIDE([Total Cost], [Total Quantity])`	DIVIDE
19	Product Sales Rank	`RANKX(ALL(Product[ProductName]), [Total Sales], , DESC)`	RANKX, ALL
20	Region Profit Rank	`RANKX(ALL(Territory[Region]), [Total Profit], , DESC)`	RANKX, ALL

5. Time-Based Analysis

Four measures drive the time analysis. Monthly Sales is the sales total in the month in context, Previous Month Sales shifts the date context back by one month with DATEADD, Sales Variance is the difference between the two, and Sales Growth % expresses that difference as a percentage of the previous month. The Calendar table must be marked as a date table for DATEADD to work correctly.

Reading the DAX Studio output for 2020, sales climbed from about 585 thousand in January to about 670 thousand in June, fell by 27.4% in July and by 35.9% in September, and then rebounded by 72.6% in December. In 2021 sales rose through the second half of the year, with a jump of 52.7% in July and more than 1.03 million in October.

Sales by calendar year, calculated from the same model, show the overall trend:

Year	Total Sales	Profit Margin	Data covers
2020	6.40M	40.6%	Jan – Dec
2021	9.32M	42.5%	Jan – Dec
2022	9.19M	42.3%	Jan – Jun only

The 2022 file ends on 30 June 2022, so 2022 is a half year. In just six months it almost matched the whole of 2021, which points to strong growth in the most recent period.

6. Validating the Measures in DAX Studio

With the model open in Power BI Desktop, I connected DAX Studio through the External tools tab, which opens it already attached to the model (the PBI / SSDT Model option on the connect screen does the same). DAX Studio queries the same in-memory model as the report, so it is a good way to check that each measure returns the right value and to test measures outside of any visual.

I ran seven queries. Each screenshot below shows the query together with its output.

Query 1: KPI check

A single ROW query returns all the headline KPIs at once, so they can be compared directly with the dashboard cards.

```
EVALUATE
ROW(
    "Total Sales", [Total Sales],
    "Total Cost", [Total Cost],
    "Total Profit", [Total Profit],
    "Total Quantity", [Total Quantity],
    "Total Orders", [Total Orders],
    "Total Customers", [Total Customers],
    "Average Order Value", [Average Order Value],
    "Profit Margin %", [Profit Margin %]
)
```

Log **Results** History

Total Sales	Total Cost	Total Profit	Total Quantity	Total Orders	Total Customers	Average Order Value	Profit Margin %
24914586.8193	14456871.3850999	10457715.4342001	84174	25164	17416	990.088492262756	41.9742679661821

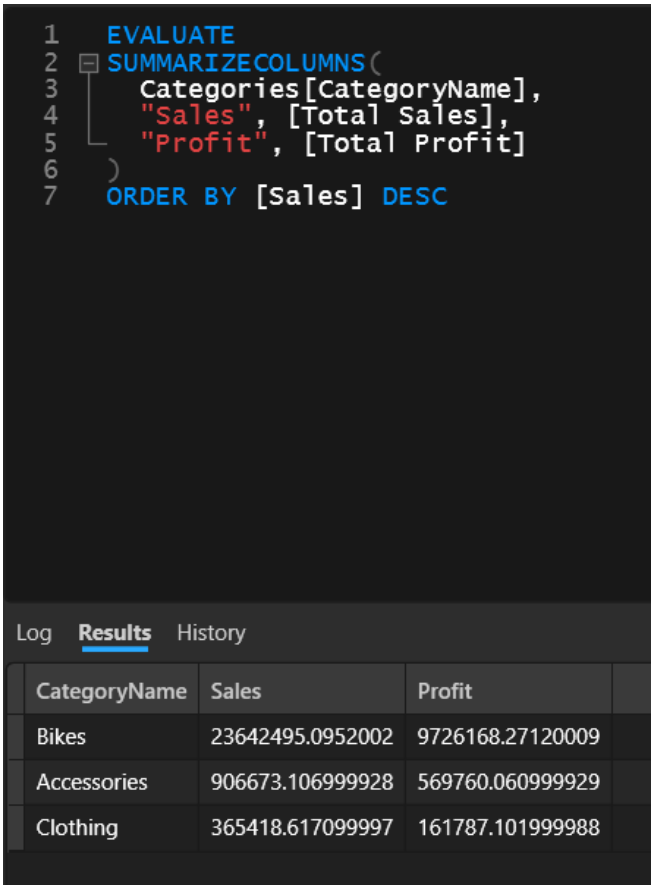
In this Query we calculate all the basic and important features from our dataset using DAX Studio.

The values returned by DAX Studio match the KPI cards on the dashboard:

KPI	DAX Studio result	Dashboard card
Total Sales	24,914,586.82	24.91M
Total Cost	14,456,871.39	not shown
Total Profit	10,457,715.43	10.46M
Total Quantity	84,174	not shown
Total Orders	25,164	25K
Total Customers	17,416	17K
Average Order Value	990.09	990.09
Profit Margin %	41.97	41.97

Query 2: Sales and profit by category

SUMMARIZECOLUMNS groups the measures by product category and ORDER BY ranks the categories by sales.



In this we are forming a table of different categories on the basis of Total Sales done and Total Profit Earned.

Query 3: Sales, profit and rank by region

This query adds the RANKX-based Region Profit Rank measure next to sales and profit for each region.

```
EVALUATE
SUMMARIZECOLUMNS(
    Territory[Region],
    "Sales", [Total Sales],
    "Profit", [Total Profit],
    "Profit Rank", [Region Profit Rank]
)
ORDER BY [Profit] DESC
```

Log **Results** History

Region	Sales	Profit	Profit Rank
Australia	7416456.20009994	3077022.81019988	1
Southwest	4822794.70059999	2040345.17109999	2
Northwest	3095074.47250003	1314751.26810002	3
United Kingdom	2902562.09490008	1214774.41580007	4
Germany	2524679.97280003	1054186.20720003	5
France	2362643.31930004	989346.05540003	6
Canada	1769245.81449999	757844.195999982	7
Southeast	11585.6164	5133.2881	8
Northeast	6401.5678	2874.5817	9
Central	3143.0604	1437.4406	10

Again in this query we are Making a table of the Territory Regions base on Total Sales, Total Profit And Profit Rank.

Query 4: Top 10 products

TOPN keeps the ten products with the highest sales, which is the same logic as the Top N filter on the dashboard bar chart.

```
EVALUATE
TOPN(
    10,
    SUMMARIZECOLUMNS(
        Product[ProductName],
        "Sales", [Total Sales],
        "Profit", [Total Profit]
    ),
    [Sales], DESC
)
ORDER BY [Sales] DESC
```

Log **Results** History

ProductName	Sales	Profit
Mountain-200 Black, 46	1241753.50919999	571632.649199986
Mountain-200 Black, 42	1233557.11639999	567859.496399987
Mountain-200 Silver, 38	1213851.8856	558788.3282
Mountain-200 Silver, 46	1182780.5916	544484.872700001
Mountain-200 Black, 38	1165936.8758	536730.985799993
Mountain-200 Silver, 42	1133066.5212	521599.343900002
Road-250 Black, 52	689373.75	272037.6692
Road-250 Red, 58	661013.4375	260846.2461
Road-250 Black, 48	641379.375	253098.3378
Road-150 Red, 48	640510.330000002	251848.668200001

In this query We are forming a table of All the Products sold again on the basis of Total Sales an Total Profit. Top sales is the Highest.

Query 5: Monthly time-based analysis

This query lists monthly sales, previous month sales, variance and growth percentage for every month, ordered by year and month number.

```
EVALUATE
SUMMARIZECOLUMNS(
    'Calendar'[Year],
    'Calendar'[Month Number],
    'Calendar'[Month Name],
    "Monthly Sales", [Total Sales],
    "Previous Month", [Previous Month Sales],
    "Variance", [Sales Variance],
    "Growth %", [Sales Growth %]
)
ORDER BY 'Calendar'[Year], 'Calendar'[Month Number]
```

Log **Results** History

Year	Month Number	Month Name	Monthly Sales	Previous Month	Variance	Growth %
2020	1	Jan	585312.648600001		585312.648600001	
2020	2	Feb	532226.245800001	585312.6486	-53086.4027999991	-9.06975151262759
2020	3	Mar	643436.104000001	532226.2458	111209.858200001	20.8952224881055
2020	4	Apr	653364.036800002	643436.104	9927.93280000158	1.54295550689235
2020	5	May	659325.896800001	653364.0368	5961.86000000127	0.912486709430908
2020	6	Jun	669988.669600001	659325.8968	10662.7728000013	1.61722341739532
2020	7	Jul	486115.0054	669988.6696	-183873.6642	-27.4442945893066
2020	8	Aug	536452.8175	486115.0054	50337.8120999997	10.3551241045479
2020	9	Sep	344062.8749	536452.8175	-192389.9426	-35.8633483363148
2020	10	Oct	404276.5974	344062.8749	60213.7225000002	17.5007903766139
2020	11	Nov	326611.1534	404276.5974	-77665.4439999997	-19.2109670704376
2020	12	Dec	563761.530100001	326611.1534	237150.376700001	72.6093932283945
2021	1	Jan	432425.7362	563761.5301	-131335.7939	-23.2963384139927

This acts as a Summary Table of Monthly Sales compared to previous months acroos all the years.

Query 6: Query-scoped measure test

A measure can be defined inside a query with DEFINE and tested without touching the model. Here I recalculated the margin as a fraction and compared it with the model measure. The test measure returned

0.4197 and the model measure returned 41.97, which is the same value expressed as a fraction and as a percentage, so the margin calculation is consistent.

```
DEFINE
    MEASURE Sales[Test Margin] = DIVIDE([Total Profit], [Total Sales])
EVALUATE
ROW("Test Margin", [Test Margin], "Profit Margin %", [Profit Margin %])
```

Log **Results** History

Test Margin	Profit Margin %
0.419742679661821	41.9742679661821

We use this query to calculate the average profit Margin acrss all the sales we have done.

Query 7: List of all measures

INFO.MEASURES returns every measure in the model with its DAX expression, which makes it a quick way to review and document the whole set.

```
EVALUATE INFO.MEASURES()
```

100 %

Log **Results** History

ID	TableID	Name	Description	DataType	Expression	FormatString	IsHidden	State	ModifiedTime	StructureModifiedTime	KPIID	IsS
4485	4458	Total Sales		8	SUMX(Sales, Sales[OrderQuantity] * RELATED(Product[ProductPrice]))		False	1	11-10-2026 13:26:15	11-10-2026 13:26:02		
4488	4458	Total Cost		8	SUMX(Sales, Sales[OrderQuantity] * RELATED(Product[ProductCost]))		False	1	11-10-2026 13:26:23	11-10-2026 13:26:19		
4491	4458	Total Profit		8	[Total Sales] - [Total Cost]		False	1	11-10-2026 13:26:29	11-10-2026 13:26:26		
4494	4458	Total Quantity		6	SUM(Sales[OrderQuantity])	0	False	1	11-10-2026 13:26:37	11-10-2026 13:26:33		
4496	4458	Total Orders		6	DISTINCTCOUNT(Sales[OrderNumber])	0	False	1	11-10-2026 13:26:45	11-10-2026 13:26:41		
4498	4458	Total Customers		6	DISTINCTCOUNT(Sales[CustomerKey])	0	False	1	11-10-2026 13:26:53	11-10-2026 13:26:49		
4500	4458	Average Sales		8	DIVIDE([Total Sales], COUNTROWS(Sales))		False	1	11-10-2026 13:27:02	11-10-2026 13:26:56		
4503	4458	Average Order Value		8	DIVIDE([Total Sales], [Total Orders])		False	1	11-10-2026 13:27:09	11-10-2026 13:27:05		
4506	4458	Average Product Price		8	AVERAGE(Product[ProductPrice])		False	1	11-10-2026 13:27:15	11-10-2026 13:27:12		
4509	4458	Profit Margin %		8	DIVIDE([Total Profit], [Total Sales]) * 100		False	1	11-10-2026 13:27:23	11-10-2026 13:27:18		
4512	4458	Monthly Sales		8	CALCULATE([Total Sales], VALUES(Calendar[Month Name]))		False	1	11-10-2026 13:27:33	11-10-2026 13:27:26		
4515	4458	Previous Month Sales		8	CALCULATE([Total Sales], DATEADD(Calendar[Date], -1, MONTH))		False	1	11-10-2026 13:27:40	11-10-2026 13:27:36		

This Query just gives us the Basic info of all The Dax Measures we have calculated. Their Data types, State, Time of Creaion/Modification & ID.

Query performance

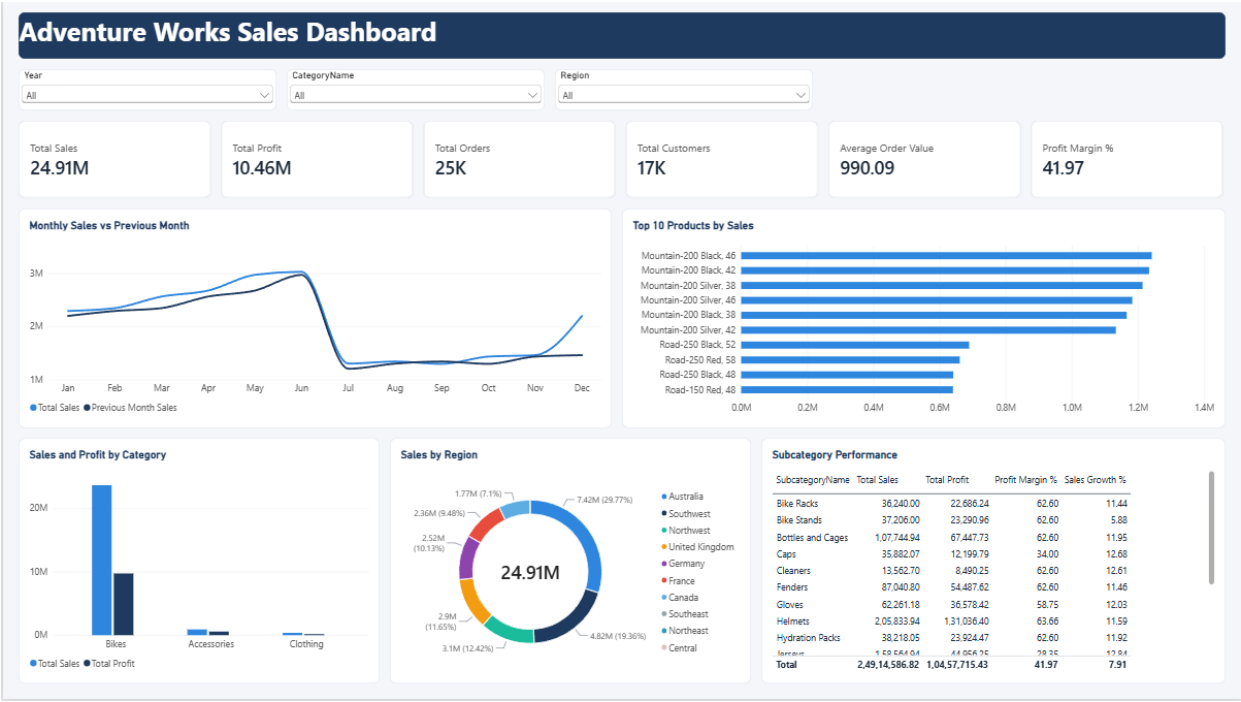
To analyse performance I switched on the Server Timings and Query Plan options in the DAX Studio ribbon and re-ran the queries. Server Timings splits the total duration into storage-engine and formula-engine time, and the query plan shows how the engine evaluates each measure. Measures built on SUMX over the whole fact table are the heaviest ones in this model, while simple aggregations such as SUM and DISTINCTCOUNT are cheap.

7. The Interactive Dashboard

The dashboard is a single page built on the same model. The top row has slicers for Year, Category and Region, followed by six KPI cards: Total Sales, Total Profit, Total Orders, Total Customers, Average Order Value and Profit Margin %.

Below the cards are the visuals: a line chart of Total Sales against Previous Month Sales by month, a bar chart of the top 10 products by sales (a Top N filter on ProductName), a column chart of sales and profit by category, a donut chart of sales by region, and a subcategory table showing Total Sales, Total Profit, Profit Margin % and Sales Growth %. Every visual responds to all three slicers, and clicking a bar or a slice cross-filters the others.

One thing to keep in mind when reading the line chart with the Year slicer on All: the 2022 data stops in June, so the second half of the year contains only 2020 and 2021, which makes the line drop after June. Selecting a single year gives a cleaner picture of the trend.



8. Analysis and Interpretation

Top-selling products

The Mountain-200 family dominates. The six best-selling products are all Mountain-200 variants, and together they bring in about 7.17 million, roughly 29% of all sales. The Road-250 and Road-150 models follow at around 640 to 690 thousand each.

Product	Sales	Profit
Mountain-200 Black, 46	1,241,754	571,633
Mountain-200 Black, 42	1,233,557	567,859
Mountain-200 Silver, 38	1,213,852	558,788
Mountain-200 Silver, 46	1,182,781	544,485
Mountain-200 Black, 38	1,165,937	536,731
Mountain-200 Silver, 42	1,133,067	521,599
Road-250 Black, 52	689,374	272,038
Road-250 Red, 58	661,013	260,846
Road-250 Black, 48	641,379	253,098
Road-150 Red, 48	640,510	251,849

Categories

Bikes are the business: 23.64 million in sales, about 95% of the total. Accessories and Clothing are small by revenue, but they are not equal in quality. Accessories earn a margin of about 63%, much higher than Bikes at about 41% and Clothing at about 44%.

Category	Sales	Profit	Margin
Bikes	23,642,495	9,726,168	41.1%
Accessories	906,673	569,760	62.8%
Clothing	365,419	161,787	44.3%

Regions

Australia is the top region with 7.42 million in sales and 3.08 million in profit, about 30% of total sales. Australia, Southwest and Northwest together account for roughly 61%. The United Kingdom, Germany, France and Canada form a middle group between 1.8 and 2.9 million. The Southeast, Northeast and Central regions are almost negligible, each under 12 thousand.

Region	Sales	Profit	Profit rank
Australia	7,416,456	3,077,023	1

Region	Sales	Profit	Profit rank
Southwest	4,822,794	2,040,345	2
Northwest	3,095,074	1,314,751	3
United Kingdom	2,902,562	1,214,775	4
Germany	2,524,680	1,054,186	5
France	2,362,643	989,346	6
Canada	1,769,246	757,844	7
Southeast	11,585	5,133	8
Northeast	6,402	2,875	9
Central	3,143	1,437	10

Sales trends

Sales grew strongly over the three years: 6.40 million in 2020, 9.32 million in 2021 (about 46% higher), and 9.19 million in the first half of 2022 alone. The first six months of 2022 are about three times the first six months of 2021. Within years, 2020 was uneven, with sharp falls in July and September followed by a December rebound, while 2021 climbed steadily from July onwards and ended with its best month in December at about 1.64 million.

Profitability

The overall profit margin is 41.97%, with total profit of 10.46 million on 24.91 million of sales. The margin is stable across the years at about 40.6% in 2020, 42.5% in 2021 and 42.3% in 2022. Margin differs more by category than by year, which makes product mix the main lever on profitability.

9. Key Insights and Conclusion

- **Revenue is concentrated.** Bikes make up about 95% of sales and the Mountain-200 line alone is about 29%, so the business depends heavily on a small set of products.
- **Accessories are the most profitable category.** At about 63% margin they are a strong candidate for growth even though they are small today.
- **Geography is uneven.** Australia, Southwest and Northwest generate about 61% of sales, while three US regions contribute almost nothing.
- **Growth is strong and margins are steady.** Sales rose sharply from 2020 to 2022 with the margin holding at about 41 to 43 percent.
- **The measures are validated.** DAX Studio returned the same values as the dashboard cards, and the query-scoped margin test agreed with the model measure.

In conclusion, the practical showed how DAX measures turn raw order lines into KPIs, how DAX Studio can be used to test and validate them independently of the report, and how an interactive dashboard makes the results easy to explore. The analysis points to Bikes, and the Mountain-200 range in particular, as the backbone of sales, with Accessories and under-served regions as the clearest opportunities.